

EduGenie

AI-Powered Learning Assistant

PROJECT TESTING & VALIDATION REPORT

This document records functional testing of the EduGenie application using the provided application screenshots. The tested features include learning-path recommendation, text summarization, quiz generation, question answering, and topic explanation.

Project	EduGenie
Purpose	AI-assisted learning and study support
Testing Type	Functional / UI feature testing
Test Cases	5 major features, 10 screenshot checkpoints
Status	Documented for project demonstration and validation

Prepared for project testing / demonstration

1. Project Overview

EduGenie is designed as an AI-powered educational assistant. The interface allows a learner to select a task, provide a topic or text, and generate an AI-assisted response.

Core features tested

1. **Summarize Text** – converts longer educational text into a simpler summary.
2. **Recommend Learning Path** – creates a step-by-step learning roadmap for a topic.
3. **Generate Quiz** – generates practice questions from a selected topic.
4. **Ask a Question** – answers a learner's direct question.
5. **Explain a Topic** – provides a beginner-friendly explanation of a concept.

Testing approach

Each feature was checked using the supplied screenshots. For every feature, the input screen and the generated output were reviewed. A successful test means the interface accepted the input and produced a relevant educational response.

Overall test summary

TC ID	Feature	Input Provided	Observed Output	Result
TC-01	Summarize Text	AI description	Structured summary	PASS
TC-02	Learning Path	Artificial Intelligence	Step-by-step roadmap	PASS
TC-03	Generate Quiz	Python basics	Quiz question(s)	PASS
TC-04	Ask a Question	What is Database	Database explanation	PASS
TC-05	Explain a Topic	Machine Learning for beginners	Beginner explanation	PASS

2. Functional Test Cases

Test Case	TC-01
Feature	Summarize Text
Test Input	Artificial intelligence is a branch of computer science...
Expected Result	The system should return a concise, structured summary covering the main ideas and applications.
Observed Result	The result screen contains a simple summary with sections for what AI is, what AI can do, and where AI is used.
Status	PASS

Test Case	TC-02
Feature	Recommend Learning Path
Test Input	Artificial Intelligence
Expected Result	The system should generate a progressive learning path from fundamentals to more advanced AI concepts.
Observed Result	The result provides stages such as beginner foundations and intermediate concepts, including Python, mathematics, and machine learning basics.
Status	PASS

Test Case	TC-03
Feature	Generate Quiz
Test Input	Python basics
Expected Result	The system should create practice questions related to Python fundamentals.
Observed Result	The result displays a Python multiple-choice question and answer options, demonstrating quiz generation.
Status	PASS

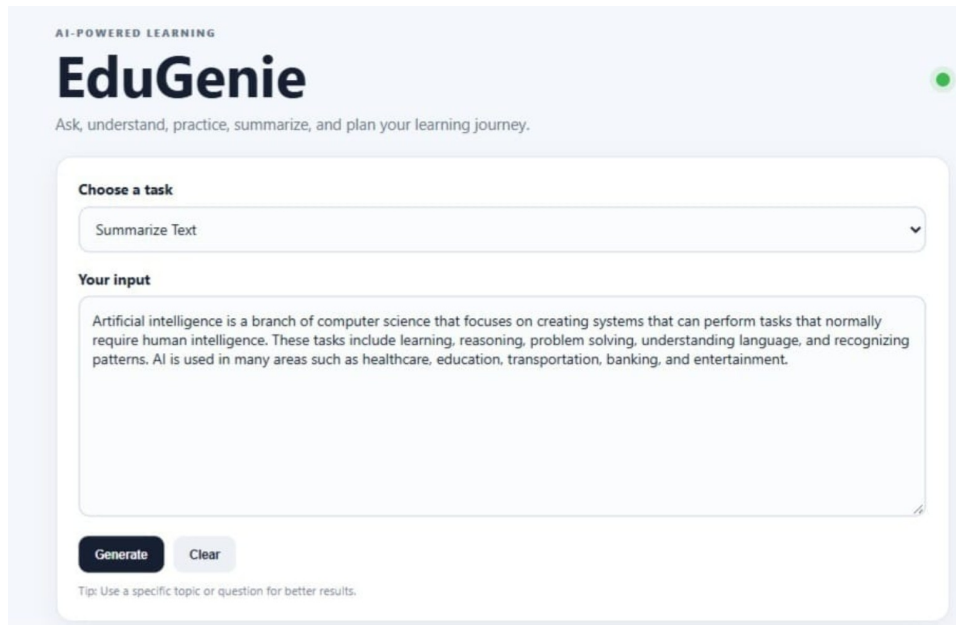
Test Case	TC-04
Feature	Ask a Question
Test Input	What is Database
Expected Result	The system should provide a direct, educational answer to the question.
Observed Result	The result explains the meaning of a database, DBMS, reasons for using databases, and database types.
Status	PASS

Test Case	TC-05
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Feature	Explain a Topic
Test Input	Explain machine learning for a beginner.
Expected Result	The system should explain machine learning in simple language suitable for a beginner.
Observed Result	The result gives a short definition and key points such as learning from data, finding patterns, and improving over time.
Status	PASS

3. Application Test Evidence

Test 1 – Summarize Text: Input Screen



The screenshot shows the 'EduGenie' interface with the heading 'AI-POWERED LEARNING'. Below the logo is the tagline 'Ask, understand, practice, summarize, and plan your learning journey.' The main section is titled 'Choose a task' and features a dropdown menu with 'Summarize Text' selected. Below this is a text area labeled 'Your input' containing a paragraph about artificial intelligence. At the bottom are 'Generate' and 'Clear' buttons, and a tip: 'Tip: Use a specific topic or question for better results.'

AI-POWERED LEARNING

EduGenie

Ask, understand, practice, summarize, and plan your learning journey.

Choose a task

Summarize Text

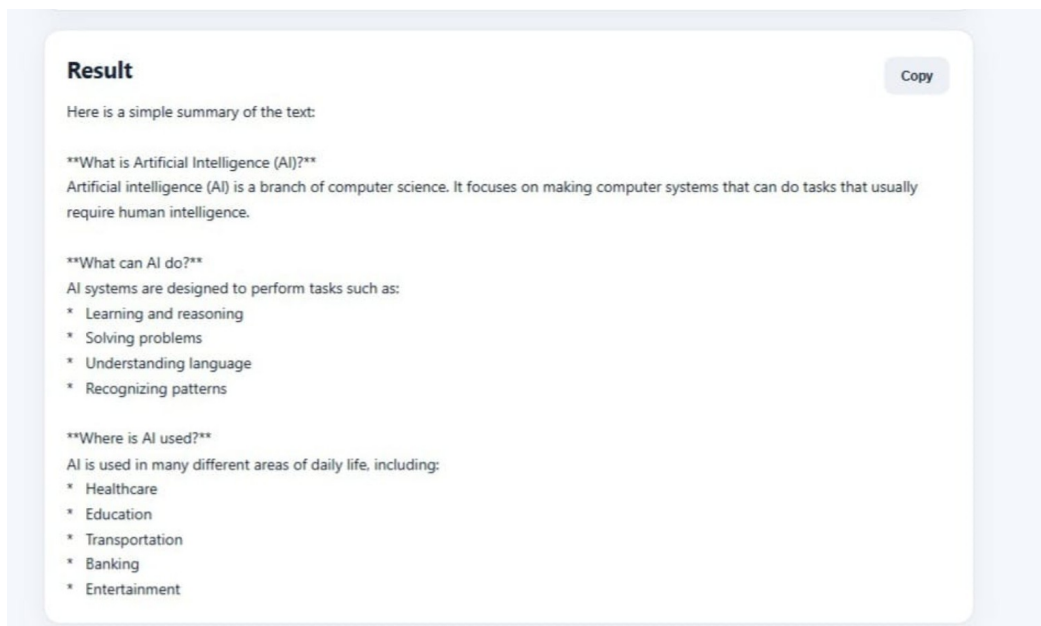
Your input

Artificial intelligence is a branch of computer science that focuses on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, understanding language, and recognizing patterns. AI is used in many areas such as healthcare, education, transportation, banking, and entertainment.

Generate **Clear**

Tip: Use a specific topic or question for better results.

Test 2 – Summarize Text: Generated Result



The screenshot shows the 'Result' section of the application. It includes a 'Copy' button in the top right corner. The text provides a simple summary of the input text, followed by three sections: 'What is Artificial Intelligence (AI)?', 'What can AI do?', and 'Where is AI used?'. Each section contains a definition or list of examples.

Result

Copy

Here is a simple summary of the text:

****What is Artificial Intelligence (AI)?****

Artificial intelligence (AI) is a branch of computer science. It focuses on making computer systems that can do tasks that usually require human intelligence.

****What can AI do?****

AI systems are designed to perform tasks such as:

- * Learning and reasoning
- * Solving problems
- * Understanding language
- * Recognizing patterns

****Where is AI used?****

AI is used in many different areas of daily life, including:

- * Healthcare
- * Education
- * Transportation
- * Banking
- * Entertainment

3. Application Test Evidence

Test 3 – Recommend Learning Path: Input Screen

AI-POWERED LEARNING

EduGenie

Ask, understand, practice, summarize, and plan your learning journey.

Choose a task

Recommend Learning Path

Your input

Artificial Intelligence

Generate **Clear**

Tip: Use a specific topic or question for better results.

Test 4 – Recommend Learning Path: Generated Result

Result Copy

Hello! I am "EduGenie", your personal learning-path designer.

Since you are a college student, I have designed this path to fit around your classes, exams, and busy schedule. We will start from the absolute basics and build up to exciting AI concepts.

Here is your personalized, step-by-step guide to learning Artificial Intelligence (AI).

1. Beginner Foundations (Months 1–2)
Goal: Build your coding and math foundations.

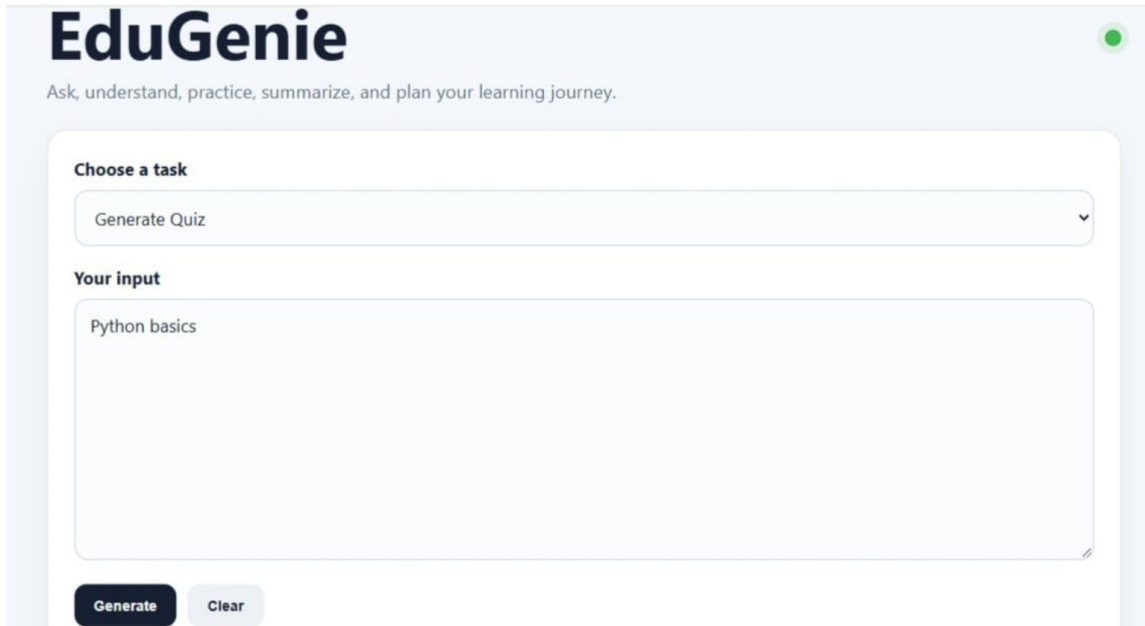
- * **Learn Python Programming:** Python is the language of AI. Focus on:
 - Variables, loops, and functions.
 - Data structures (lists, dictionaries).
 - Libraries for handling data: **NumPy** (for math) and **Pandas** (for tables/dataframes).
- * **Essential Math:** You do not need a math degree, but you need the basics:
 - Linear Algebra:** Matrices and vectors.
 - Calculus:** Derivatives (how things change).
 - Probability & Statistics:** Mean, median, mode, and probability distributions.
- * **Basic AI Concepts:** Learn the difference between AI, Machine Learning (ML), and Deep Learning (DL).

2. Intermediate Concepts (Months 3–4)
Goal: Learn how computers learn from data.

- * **Machine Learning (ML) Basics:**
 - Supervised Learning:** Teaching the computer with labeled data (e.g., predicting house prices).
 - Unsupervised Learning:** Letting the computer find patterns on its own (e.g., grouping customers by shopping habits).

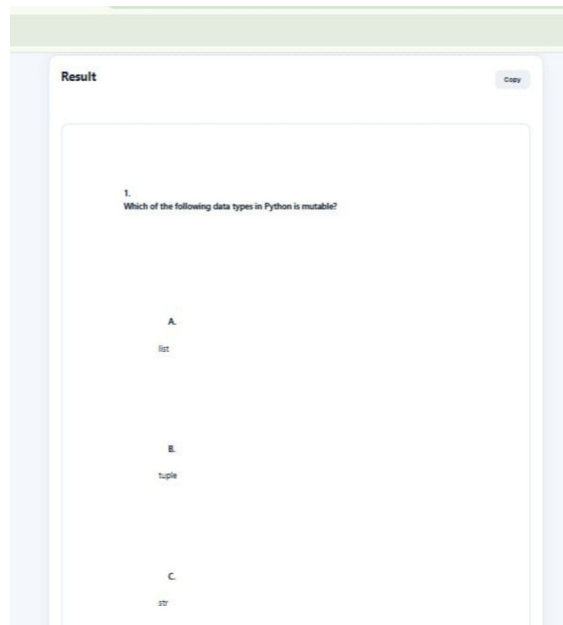
3. Application Test Evidence

Test 5 – Generate Quiz: Input Screen



The screenshot shows the EduGenie application interface. At the top, the logo 'EduGenie' is displayed in a large, bold, dark blue font. Below the logo, a tagline reads: 'Ask, understand, practice, summarize, and plan your learning journey.' The main content area is a light blue box with a white border. It contains a section titled 'Choose a task' with a dropdown menu showing 'Generate Quiz'. Below this is a section titled 'Your input' with a large text area containing the text 'Python basics'. At the bottom of the input section are two buttons: 'Generate' (dark blue) and 'Clear' (light blue).

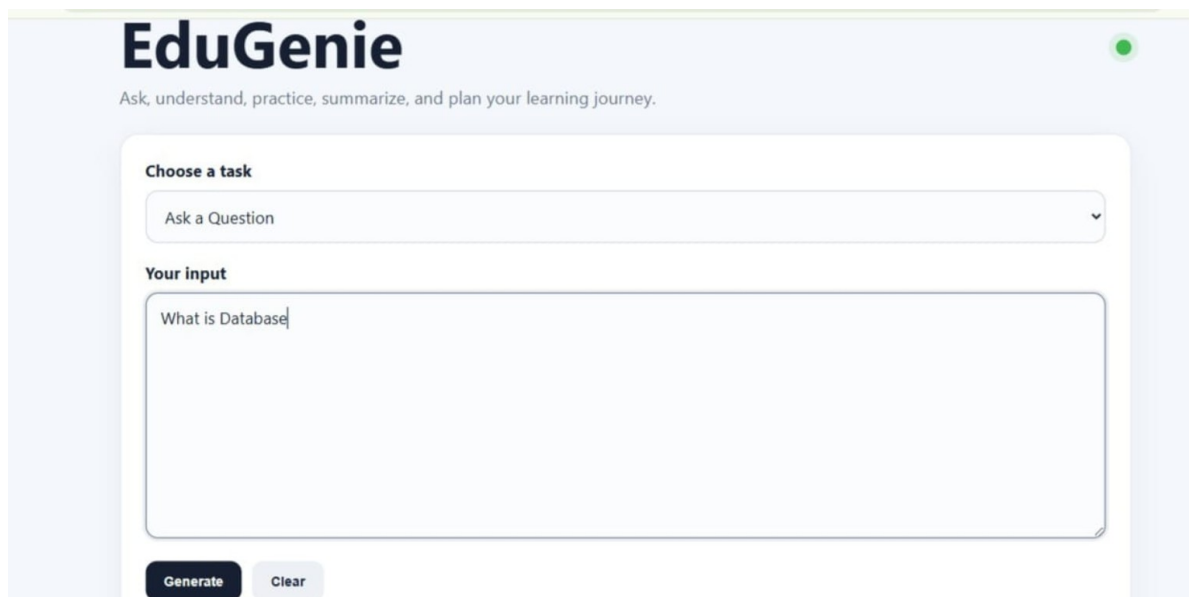
Test 6 – Generate Quiz: Generated Result



The screenshot shows the 'Result' screen of the EduGenie application. The title 'Result' is at the top left, and a 'Copy' button is at the top right. The main content area is a white box with a light blue border. It contains a single question: '1. Which of the following data types in Python is mutable?'. Below the question are three options: 'A. list', 'B. tuple', and 'C. str'.

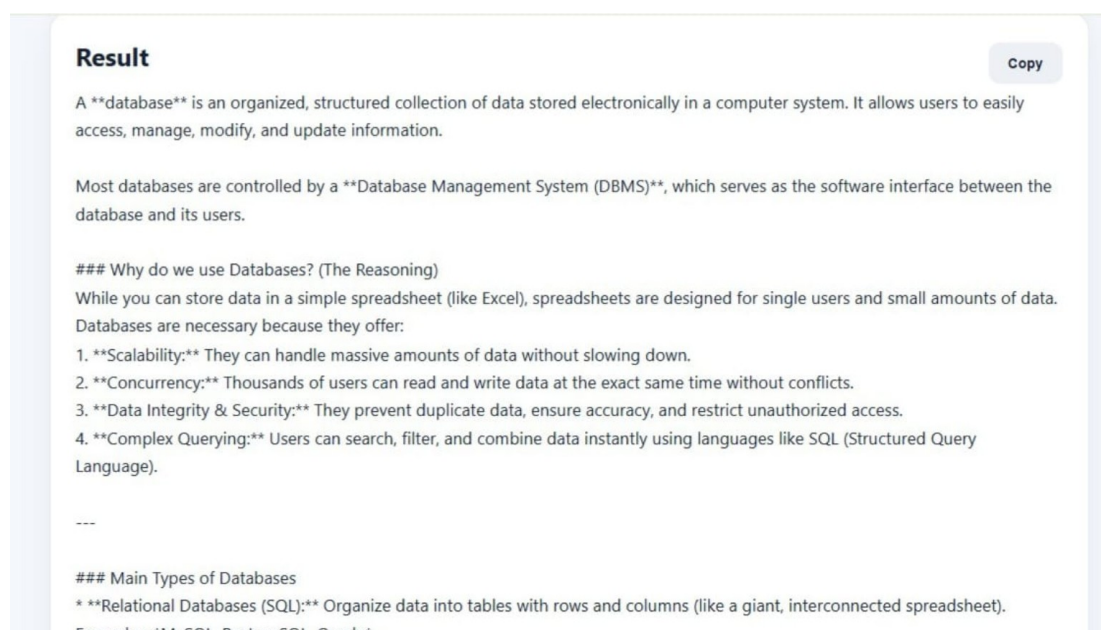
3. Application Test Evidence

Test 7 – Ask a Question: Input Screen



The screenshot shows the EduGenie application interface. At the top, the logo 'EduGenie' is displayed with the tagline 'Ask, understand, practice, summarize, and plan your learning journey.' Below this, there is a section titled 'Choose a task' with a dropdown menu currently set to 'Ask a Question'. Underneath, the 'Your input' section contains a large text area where the user has typed 'What is Database'. At the bottom of the input section, there are two buttons: 'Generate' (in dark blue) and 'Clear' (in light blue).

Test 8 – Ask a Question: Generated Result



The screenshot displays the 'Result' section of the application. It features a 'Copy' button in the top right corner. The main content area contains the following text:

A **database** is an organized, structured collection of data stored electronically in a computer system. It allows users to easily access, manage, modify, and update information.

Most databases are controlled by a **Database Management System (DBMS)**, which serves as the software interface between the database and its users.

Why do we use Databases? (The Reasoning)

While you can store data in a simple spreadsheet (like Excel), spreadsheets are designed for single users and small amounts of data. Databases are necessary because they offer:

1. **Scalability:** They can handle massive amounts of data without slowing down.
2. **Concurrency:** Thousands of users can read and write data at the exact same time without conflicts.
3. **Data Integrity & Security:** They prevent duplicate data, ensure accuracy, and restrict unauthorized access.
4. **Complex Querying:** Users can search, filter, and combine data instantly using languages like SQL (Structured Query Language).

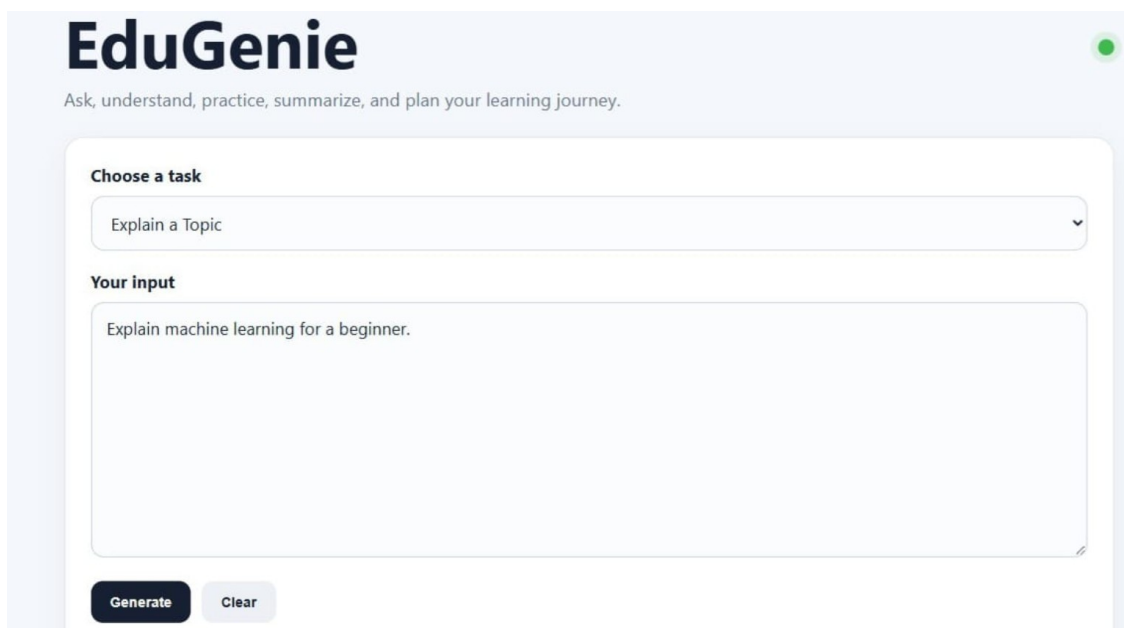
Main Types of Databases

* **Relational Databases (SQL):** Organize data into tables with rows and columns (like a giant, interconnected spreadsheet).

Examples: *MySQL *PostgreSQL *Oracle*

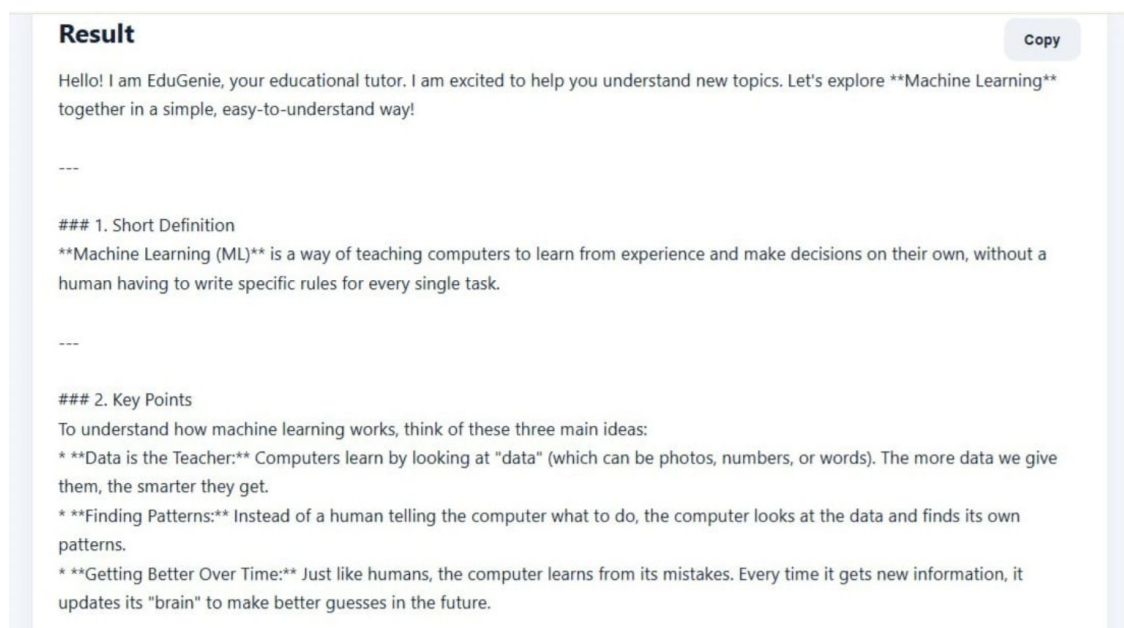
3. Application Test Evidence

Test 9 – Explain a Topic: Input Screen



The screenshot shows the EduGenie application interface. At the top, the logo "EduGenie" is displayed in a large, bold, dark blue font. Below the logo, a tagline reads: "Ask, understand, practice, summarize, and plan your learning journey." The main content area is a light blue rounded rectangle. Inside, there's a section titled "Choose a task" with a dropdown menu currently showing "Explain a Topic". Below this is a section titled "Your input" with a large text area containing the text "Explain machine learning for a beginner." At the bottom of the input section are two buttons: "Generate" (dark blue) and "Clear" (light blue).

Test 10 – Explain a Topic: Generated Result



The screenshot shows the "Result" section of the EduGenie application. The word "Result" is in bold at the top left, and a "Copy" button is at the top right. The main text area contains the following content:

Hello! I am EduGenie, your educational tutor. I am excited to help you understand new topics. Let's explore **Machine Learning** together in a simple, easy-to-understand way!

1. Short Definition

Machine Learning (ML) is a way of teaching computers to learn from experience and make decisions on their own, without a human having to write specific rules for every single task.

2. Key Points

To understand how machine learning works, think of these three main ideas:

- Data is the Teacher:** Computers learn by looking at "data" (which can be photos, numbers, or words). The more data we give them, the smarter they get.
- Finding Patterns:** Instead of a human telling the computer what to do, the computer looks at the data and finds its own patterns.
- Getting Better Over Time:** Just like humans, the computer learns from its mistakes. Every time it gets new information, it updates its "brain" to make better guesses in the future.

4. Testing Conclusion

Based on the supplied screenshots, all five demonstrated EduGenie features produced visible outputs for their corresponding inputs. The evidence shows that the application can support multiple learning activities through a single interface.

Validated functional areas

- User task selection and input handling
- AI-generated educational responses
- Learning-path generation
- Text summarization
- Quiz generation
- Question answering
- Beginner-friendly topic explanation

Future testing recommendations

For a complete software-testing submission, the next stage can include validation of empty inputs, very long inputs, invalid characters, API/network failures, loading states, copy-button behavior, repeated requests, and mobile/responsive layouts.

Final status: The provided screenshot evidence demonstrates successful functional operation of the main EduGenie learning features.